

Exercise #2

23. August 2023

Problem 1. (Polynomial interpolation)

You are allowed (but not required) to use python for this problem. Approximate answers obtained by numerical methods are accepted. Consider the function

$$f(x) = 3^{-(x-1)^2}.$$

- a) Interpolate $f(x)$ by a polynomial of minimal degree, by using the interpolation points

$$x_0 = -1, \quad x_1 = 0, \quad x_2 = 1, \quad x_3 = 2$$

What is the maximal error on the intervals $[-1, 2]$ and $[-5, 5]$? Plot $f(x)$ and the interpolating polynomial in the same plot, on the interval $[-1, 2]$.

- b) The *Gaussian nodes* on the interval $[-1, 1]$ are defined, for $i = 0, \dots, n$, as

$$\tilde{x}_0 = \sqrt{\frac{3}{5} - \frac{2}{7}\sqrt{\frac{6}{5}}}, \quad \tilde{x}_1 = -\sqrt{\frac{3}{5} - \frac{2}{7}\sqrt{\frac{6}{5}}}, \quad \tilde{x}_2 = \sqrt{\frac{3}{5} + \frac{2}{7}\sqrt{\frac{6}{5}}}, \quad \tilde{x}_3 = -\sqrt{\frac{3}{5} + \frac{2}{7}\sqrt{\frac{6}{5}}}$$

and are transported on a general interval $[a, b]$ with the transformation rule:

$$x_i = \frac{1}{2}(a+b) + \frac{1}{2}(b-a)\tilde{x}_i.$$

Find the Gaussian nodes on the interval $[-1, 2]$ for $n = 3$.

- c) Interpolate $f(x)$ in the Gaussian nodes for $n = 3$. What is the maximal error on the intervals $[-1, 2]$ and $[-5, 5]$? Plot $f(x)$ and the interpolating polynomial in the same plot, on the interval $[-1, 2]$.
- d) Plot the error as a function in x , for both the interpolation in a) and the interpolation in c), on the interval $[-1, 2]$. Plot both errors in the same plot.

Solution.

- a) Evaluating the function gives the data points

i	0	1	2	3
x_i	-1	0	1	2
y_i	$\frac{1}{81}$	$\frac{1}{3}$	1	$\frac{1}{3}$

so that we get the divided difference table

-1	$\frac{1}{81}$			
		$\frac{26}{81}$		
0	$\frac{1}{3}$		$\frac{14}{81}$	
		$\frac{2}{3}$		$-\frac{68}{243}$
1	1		$-\frac{2}{3}$	
		$-\frac{2}{3}$		
2	$\frac{1}{3}$			

and interpolating polynomial

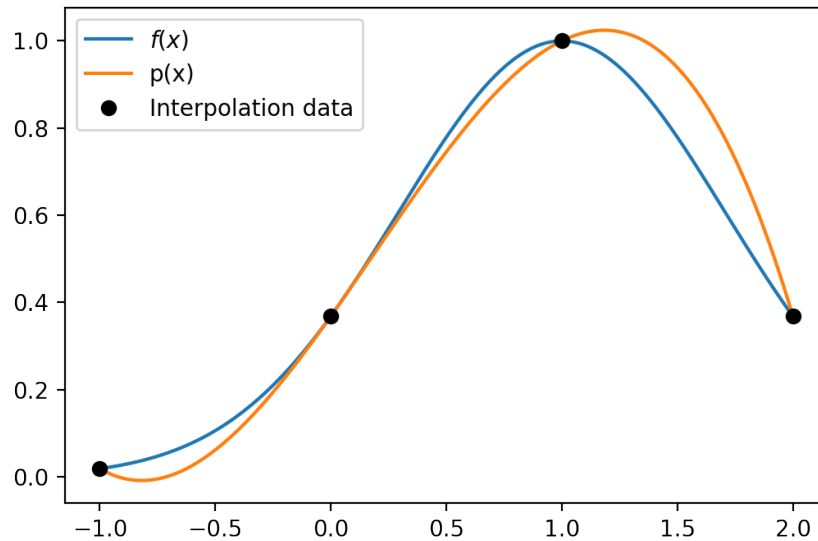
$$\begin{aligned}
 & \frac{1}{81} + \frac{26}{81}(x+1) + \frac{14}{81}(x+1)(x-0) - \frac{68}{243}(x+1)(x-0)(x-1) \\
 &= \frac{1}{81} \frac{(x-0)(x-1)(x-2)}{((-1)-0)((-1)-1)((-1)-2)} + \frac{1}{3} \frac{(x-(-1))(x-1)(x-2)}{(0-(-1))(0-1)(0-2)} \\
 &+ 1 \frac{(x-(-1))(x-0)(x-2)}{(1-(-1))(1-0)(1-2)} + \frac{1}{3} \frac{(x-(-1))(x-0)(x-1)}{(2-(-1))(2-0)(2-1)} \\
 &= -\frac{68}{243}x^3 + \frac{14}{81}x^2 + \frac{188}{243}x + \frac{1}{3}.
 \end{aligned}$$

The maximal error

$$\max\left\{\left|f(x) + \frac{68}{243}x^3 - \frac{14}{81}x^2 - \frac{188}{243}x - \frac{1}{3}\right| : x \in [-1, 2]\right\}$$

is approximately 0.175.

Plot:



b) For $n = 3$, we get in the interval $[-1, 2]$ the following gaussian nodes:

$$\begin{aligned} x_0 &\approx 1.792, \\ x_1 &\approx 1.01, \\ x_2 &\approx -0.01, \\ x_3 &\approx -0.792. \end{aligned}$$

c) Interpolating in these (approximated) points gives the polynomial

$$-0.309294x^3 + 0.184966x^2 + 0.754262x + 0.368082.$$

The maximal error

$$\max\{|f(x) - (-0.309294x^3 + 0.184966x^2 + 0.754262x + 0.368082)| : x \in [-1, 2]\}$$

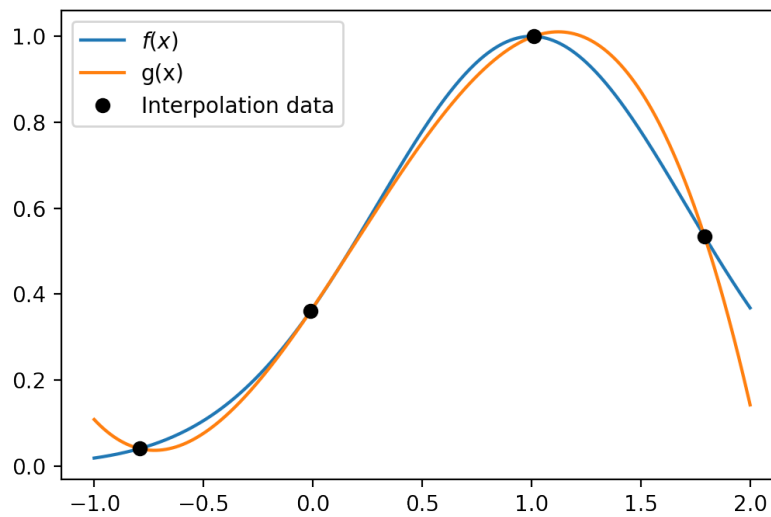
is approximately 0.226.

The maximal error

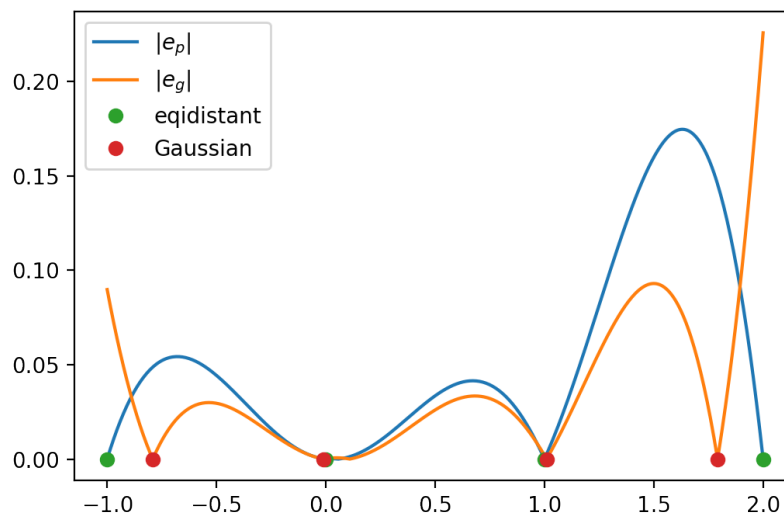
$$\max\{|f(x) - (-0.309294x^3 + 0.184966x^2 + 0.754262x + 0.368082)| : x \in [-5, 5]\}$$

is approximately 40.

Plot:



d) Errors plot:



Problem 2. (Lagrange interpolation)

In the lectures and in Maths 3, you have seen how interpolation polynomials can be constructed by solving a system of linear equations. In this exercise, we will discuss an alternative, direct approach, called *Lagrange interpolation*.

(See the note `Introduction.pdf`, Section 2.1 for more information about that approach.)

Given $n + 1$ distinct points x_i , $i = 0, \dots, n$, we define the associated *cardinal functions* as

$$\ell_i(x) = \prod_{\substack{j=0 \\ j \neq i}}^n \frac{x - x_j}{x_i - x_j} = \frac{x - x_0}{x_i - x_0} \dots \frac{x - x_{i-1}}{x_i - x_{i-1}} \cdot \frac{x - x_{i+1}}{x_i - x_{i+1}} \dots \frac{x - x_n}{x_i - x_n}, \quad i = 0, \dots, n.$$

a) Compute the cardinal functions $\ell_i(x)$ explicitly for the points $x_0 = -1$, $x_1 = 1$, $x_2 = 5$.

b) Verify that the cardinal functions satisfy the equations

$$\begin{aligned} \ell_i(x_i) &= 1 && \text{for all } i = 0, \dots, n, \\ \ell_i(x_j) &= 0 && \text{for all } i, j = 0, \dots, n \text{ with } i \neq j. \end{aligned}$$

c) Assume we are given points $(x_i, y_i)_{i=0}^n$, with distinct values x_i . Show that the interpolation polynomial p_n is given by

$$p_n(x) = \sum_{i=0}^n y_i \ell_i(x).$$

(Hint: Verify that each ℓ_i is a polynomial of degree n and that the interpolation condition $p_n(x_i) = y_i$ is satisfied.)

Solution.

a) The cardinal functions are

$$\begin{aligned}\ell_0 &= \frac{(x-1)(x-5)}{((-1)-1)((-1)-5)} = \frac{x^2}{12} - \frac{x}{2} + \frac{5}{12}, \\ \ell_1 &= \frac{(x-(-1))(x-5)}{(1-(-1))(1-5)} = -\frac{x^2}{8} + \frac{x}{2} + \frac{5}{8}, \\ \ell_2 &= \frac{(x-(-1))(x-1)}{(5-(-1))(5-1)} = \frac{x^2}{24} - \frac{1}{24}.\end{aligned}$$

b) The cardinal functions satisfy the equation.

$$\ell_i(x_i) = 1, \quad \text{for all } i = 0, \dots, n,$$

because replacing $x = x_i$

$$\ell_i(x_i) = \prod_{\substack{j=0 \\ j \neq i}}^n \frac{x_i - x_j}{x_i - x_j} = \prod_{j=0, j \neq i}^n 1 = 1$$

and similarly to the equation

$$\ell_i(x_j) = 0 \text{ for all } i, j = 0, \dots, n \text{ with } i \neq j,$$

replacing $x = x_j$

$$\ell_i(x_j) = \prod_{k=0, k \neq i}^n \frac{x_j - x_k}{x_i - x_k} = 0$$

since there is a factor 0 when $k = j$.

c) Given the expression for the cardinal functions, we can notice that ℓ_i is the product of n linear polynomials of the form $(x - x_j)/(x_i - x_j)$. Hence ℓ_i is a polynomial of degree n . Moreover, by using the part b), we have that

$$p_n(x_i) = \sum_{j=1}^n y_j \ell_j(x_i) = y_i \ell_i(x_i) = y_i,$$

since $\ell_j(x_i) = 0$ for any $i \neq j$.

Problem 3. (Trapezoidal rule (+ adaptivity))

We consider in this problem the numerical integration of the function $f(x) = \sqrt{x}$ on the interval $I = [0, 2]$.

- a) Implement the composite trapezoidal rule for the numerical computation of this integral.
- b) Estimate the numerical converge rate for this method. That is, assume that the error satisfies

$$\text{err}_m := \left| T_m(0, 1) - \int_0^1 \sqrt{x} dx \right| \approx Ch^p,$$

and estimate numerically the number p . (Your estimated p is not necessarily an integer!)

- c) Explain why your result from part b) does not contradict the usual theory of the trapezoidal rule.
- d) Suppose an adaptive trapezoidal rule is implemented. Then, for every sub-interval $[x_i, x_{i+1}]$ two estimates are compared: the estimate for the single interval and the one for a (possible) splitted interval. They are, respectively:

$$T_1 = \frac{x_{i+1} - x_i}{2} (f(x_i) + f(x_{i+1})), \quad T_2 = \frac{x_{i+1} - x_i}{4} \left(f(x_i) + 2f\left(\frac{x_i + x_{i+1}}{2}\right) + f(x_{i+1}) \right)$$

The estimated error difference is computed as $\epsilon = |T_1 - T_2|/3$. If $\epsilon > \text{tol}$, where tol is a desired tolerance, the interval is splitted and the procedure repeated in every sub-interval until the criteria $\epsilon < \text{tol}$ is satisfied. Suppose an initial uniform grid with $m = 10$ (9 intervals) and check (manually or numerically) which intervals would need at least a first refinement with $\text{tol} = 10^{-4}$.

Solution.

- a) The composite trapezoidal rule is given by

$$\int_a^b f(x) dx = \sum_{j=0}^{m-1} T(X_j, X_{j+1}) = \sum_{j=0}^{m-1} \frac{h}{2} [f(X_j) + f(X_{j+1})] = \frac{h}{2} \left[f(a) + 2 \sum_{j=1}^{m-1} f(X_j) + f(b) \right]$$

The (approximated) numerical value of the integral considering $m = 20$ is: 1.878845

- b) The exact value of the integral is

$$\int_0^2 \sqrt{x} dx = \frac{4\sqrt{2}}{3}$$

Using the numerical verification shown in *Preliminaries*,

$$p \approx \frac{\log(e(h_{m+1})/e(h_m))}{\log(h_{m+1}/h_m)}$$

We choose a large number for m , for example, $m = 400$, then the value of p is approximately 1.49

- c) The value of p is different from 2. From the usual theory, we would expect that $p = 2$ if f is 2-times continuously differentiable. However, the first derivative of the function $f(x) = \sqrt{x}$ in $x = 0$ does not exist.

- d) A numerical check reveals that the errors in every sub interval are:

$$\epsilon_1 \approx 3.62e - 03$$

$$\epsilon_2 \approx 3.08e - 04$$

$$\epsilon_3 \approx 1.40e - 04$$

$$\epsilon_4 \approx 8.39e - 05$$

$$\epsilon_5 \approx 5.74e - 05$$

$$\epsilon_6 \approx 4.24e - 05$$

$$\epsilon_7 \approx 3.30e - 05$$

$$\epsilon_8 \approx 2.66e - 05$$

$$\epsilon_9 \approx 2.20e - 05$$

thus the first three intervals are in need of at least 1 refinement.

Problem 4. (Runge phenomenon)

You are allowed (but not required) to use python for this problem. Approximate answers obtained by numerical methods are accepted. Consider the function

$$f(x) = \frac{1}{1 + 25x^2}.$$

- a) Interpolate $f(x)$ by a polynomial of minimal degree, by using the interpolation points

$$x_0 = -1, \quad x_1 = 0, \quad x_2 = 1$$

What is the maximal error on the intervals $[-1, 1]$? Plot $f(x)$ and the interpolating polynomial in the same plot, on the interval $[-1, 1]$.

b) The *Chebyshev nodes* on an interval $[a, b]$ are defined as

$$\tilde{x}_i = \frac{a+b}{2} + \frac{b-a}{2} \cos\left(\frac{(2i+1)\pi}{2(n+1)}\right), \quad i = 0, \dots, n.$$

Interpolate $f(x)$ in the Chebyshev nodes for $n = 2$. What is the maximal error on the interval $[-1, 1]$? Plot $f(x)$ and the interpolating polynomial in the same plot, on the interval $[-1, 1]$.

- c) Repeat the analysis of a) and b) (numerically) for $n = 15$ and plot the results.
- d) Plot also the two errors in the same graph and observe the Runge phenomenon for high order degree polynomials and how the Chebyshev nodes can solve the problem.

Solution.

a) Evaluating the function gives the data points

i	0	1	2
x_i	-1	0	1
y_i	$\frac{1}{26}$	1	$\frac{1}{26}$

so that we get the divided difference table

-1	$\frac{1}{26}$		
		$\frac{25}{26}$	
0	1		$-\frac{25}{26}$
		$-\frac{25}{26}$	
1	$\frac{1}{26}$		

and interpolating polynomial

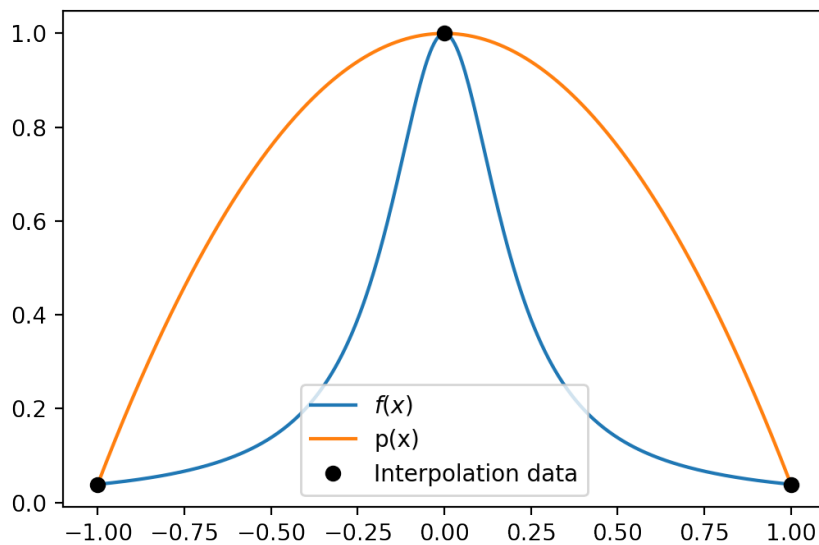
$$\begin{aligned}
 & \frac{1}{26} + \frac{25}{26}(x+1) - \frac{25}{26}(x+1)(x-0) \\
 &= \frac{1}{26} \frac{(x-0)(x-1)}{((-1)-0)((-1)-1)} + \frac{(x-(-1))(x-1)}{(0-(-1))(0-1)} \\
 &+ \frac{1}{26} \frac{(x-(-1))(x-0)}{(1-(-1))(1-0)} \\
 &= -\frac{25}{26}x^2 + 1.
 \end{aligned}$$

The maximal error

$$\max\left\{\left|f(x) + \frac{25}{26}x^2 - 1\right| : x \in [-1, 1]\right\}$$

is approximately 0.646.

Plot:



b) The Chebyshev nodes on $[-1, 1]$ are given by

$$\hat{x}_i = \cos\left(\frac{2i+1}{2n+2}\pi\right), \quad i = 0, \dots, n.$$

For $n = 2$, we get:

$$\hat{x}_0 = \cos\left(\frac{1}{6}\pi\right) = \frac{\sqrt{3}}{2},$$

$$\hat{x}_1 = \cos\left(\frac{1}{2}\pi\right) = 0,$$

$$\hat{x}_2 = \cos\left(\frac{5}{6}\pi\right) = -\frac{\sqrt{3}}{2}.$$

Luckily, there is no need to transfer them since the interval is already the reference one. Interpolating in these points gives the polynomial

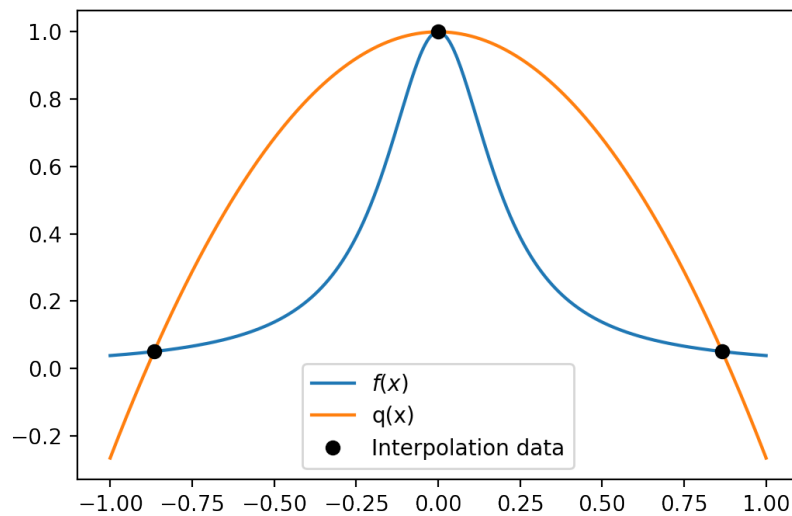
$$-1.28205x^2 + 1.$$

The maximal error

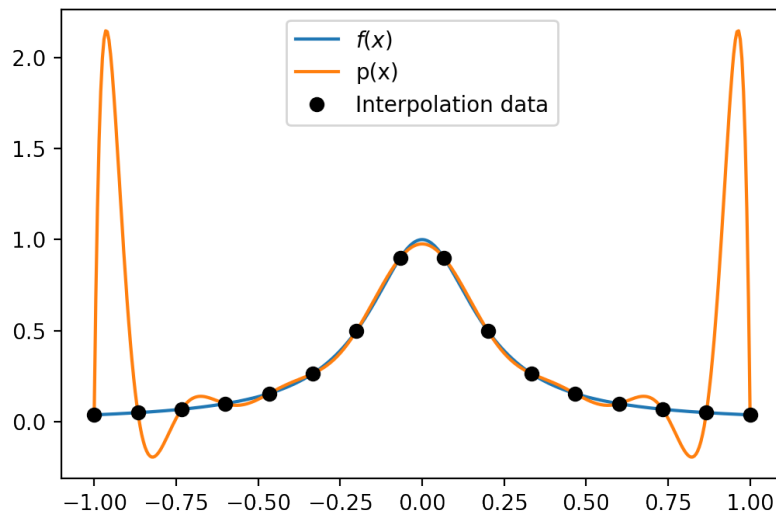
$$\max\{|f(x) - (-1.28205x^2 + 1)| : x \in [-1, 1]\}$$

is approximately 0.601.

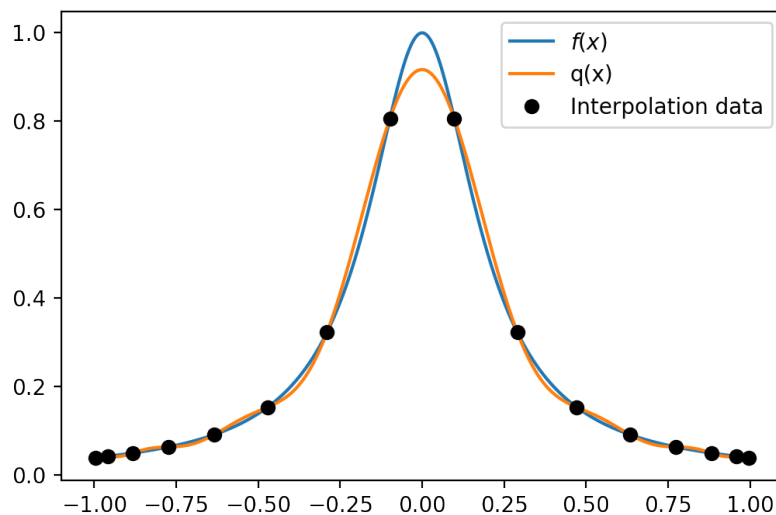
Plot:



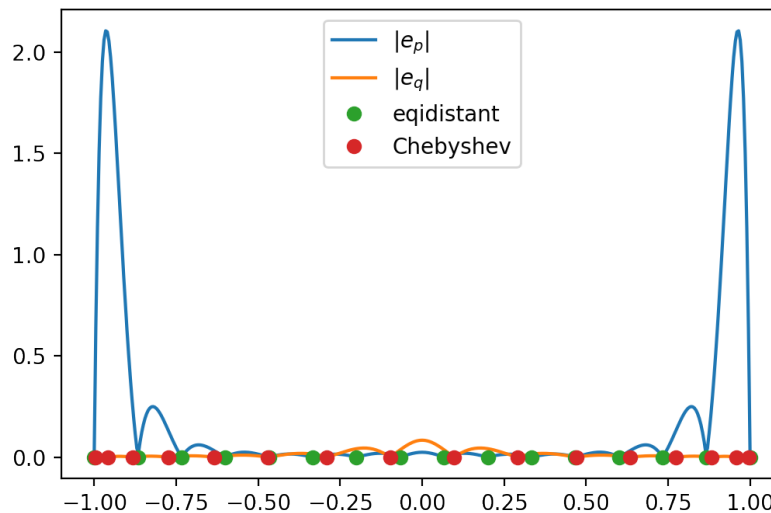
c) Plot for polynomial interpolation:



Plot for Chebyshev nodes:



d) Error plot:



Problem 5.

We consider the cubic splines $\mathcal{S}_3$ with knot vector $\Delta = [-2, -1, 0, 1, 2]$.

Which of the following functions is in the space $\mathcal{S}_{3,\Delta}$?

State a reason for each of the functions.

a) $f_1(x) = 2x$,

b) $f_2(x) = |x - 1|^3 + |x + 1|^3$,

c) $f_3(x) = 3(x)_+^2 + (x + 1)_+^3$, where $(y)_+ = \begin{cases} y & \text{if } y \geq 0 \\ 0 & \text{else.} \end{cases}$

d) For some $a, b \in \mathbb{R}$ we consider the function $f_{a,b}(x) = \begin{cases} x^3 + x & \text{for } x \in [-2, 0], \\ x^3 - ax + b & \text{for } x \in [0, 2]. \end{cases}$

Determine for which $a, b \in \mathbb{R}$ this function is a spline in the vector space $\mathcal{S}_{3,\Delta}$ and for which it is not.

Solution.

We have to check whether for a function f

1. $f|_{I_j} \in \mathbb{P}_3, I_j = [x_{j-1}, x_j],$
2. $f \in C^2[a, b],$

where we here have $x_j = -2 + j, j = 0, \dots, 4.$ and $a = x_0 = -2, b = x_4 = 4.$ We obtain

- a) $f_1(x)$ is a linear function on the whole interval $[-2, 2]$ and hence a spline.
- b) We first eliminate the absolute values by considering the separate cases

$$\begin{aligned}
 f_2(x) &= \begin{cases} (-(x-1))^3 + (-(x+1))^3 & \text{for } x \in [-2, -1), \\ (-(x-1))^3 + (x+1)^3 & \text{for } x \in [-1, 1), \\ (x-1)^3 + (x+1)^3 & \text{for } x \in [1, 2] \end{cases} \\
 &= \begin{cases} -(x-1)^3 - (x+1)^3 & \text{for } x \in [-2, -1), \\ -(x-1)^3 + (x+1)^3 & \text{for } x \in [-1, 1), \\ (x-1)^3 + (x+1)^3 & \text{for } x \in [1, 2]. \end{cases}
 \end{aligned}$$

We have a polynomial of degree at most 3 and the first part holds. Furthermore $f_2^-(-1) = 8 = f_2^+(-1)$ and $f_2^-(1) = 8 = f_2^+(1)$ so the function is continuous on $[-2, 2]$. We further get

$$f_2'(x) = \begin{cases} -3(x-1)^2 - 3(x+1)^2 & \text{for } x \in [-2, -1), \\ -3(x-1)^2 + 3(x+1)^2 & \text{for } x \in [-1, 1), \\ 3(x-1)^2 + 3(x+1)^2 & \text{for } x \in [1, 2] \end{cases}$$

checking again the left and right limits at $x_1 = -1$ and $x_3 = 1$, we see that the function f_2 is continuously differentiable on $[-2, 2]$. Finally we check

$$f_2''(x) = \begin{cases} -6(x-1) - 6(x+1) & \text{for } x \in [-2, -1), \\ -6(x-1) + 6(x+1) & \text{for } x \in [-1, 1), \\ 6(x-1) + 6(x+1) & \text{for } x \in [1, 2]. \end{cases}$$

which is continuous as well and we obtain a spline in $\mathcal{S}_3$.

c) We first resolve the definitions of $(\cdot)_+$ and obtain

$$f_3(x) = \begin{cases} 0 & \text{for } x \in [-2, -1), \\ (x+1)^3 & \text{for } x \in [-1, 0), \\ 3x^2 + (x+1)^3 & \text{for } x \in [0, 1), \\ 3x^2 + (x+1)^3 & \text{for } x \in [1, 2] \end{cases}$$

and we see that in every segment we have a polynomial of degree at most 3, so 1. is fulfilled

Checking the points $-1, 0, 1$ we see that the function is continuous

We compute the first and second derivative

$$f'_3(x) = \begin{cases} 0 & \text{for } x \in [-2, -1), \\ 3(x+1)^2 & \text{for } x \in [-1, 0), \\ 6x + 3(x+1)^2 & \text{for } x \in [0, 1), \\ 6x + 3(x+1)^2 & \text{for } x \in [1, 2]. \end{cases} \quad \text{and} \quad f''_3(x) = \begin{cases} 0 & \text{for } x \in [-2, -1), \\ 6(x+1) & \text{for } x \in [-1, 0), \\ 6 + 6(x+1) & \text{for } x \in [0, 1), \\ 6 + 6(x+1) & \text{for } x \in [1, 2]. \end{cases}$$

The first derivative is still continuous, but the second is not, since

$$f''_3(0)|_{[-1,0]} = 6 \neq 12 = f''_3(0)|_{[0,1]}.$$

and we do not have a spline in this case.

c) If $f_{a,b}(x)$ is a spline, it has to be continuous in $x = 0$ and we obtain that $b = 0$.

For the first derivative we obtain

$$f'_{a,b}(x) = \begin{cases} 3x^2 + 1 & \text{for } x \in [-2, 0), \\ 3x^2 - a & \text{for } x \in [0, 2]. \end{cases}$$

the first derivative has to be continuous at $x = 0$ as well and we obtain that in order to $f_{a,b}$ being a spline we need $a = -1$.

We continue

$$f''_{a,b}(x) = \begin{cases} 6x & \text{for } x \in [-2, 0), \\ 6x & \text{for } x \in [0, 2]. \end{cases}$$

which is continuous.

Hence $f_{a,b}(x)$ is a spline if and only if $a = -1$ und $b = 0$.

The next exercises are optional and should not be handed in! Problem 6. (Inverse interpolation)

We are given a function $f(x)$ with a root r in an interval $[a, b]$, that is, $f(r) = 0$. Moreover, we assume that the interval is chosen sufficiently small such that the function f is either increasing on the whole interval, or decreasing on the whole interval. We want to find r by using interpolation. To that end, we compute an interpolating function $p(y)$ through the points $(f(x_i), x_i)_{i=0}^n$, for nodes $x_i \in [a, b]$. Then $p(0)$ should be an approximation of r .

- a) Let $f(x) = x^3 - 7$ and $[a, b] = [1.5, 2.0]$. Choose the nodes $x_i = (1.5, 1.75, 2.0)$ and compute an interpolation polynomial in order to approximate the root r . How close are you to the actual root?
- b) Repeat the example with $n + 1$ equidistant nodes over the interval $[a, b]$. Choose $n = 2, 4, 8$, and 16 , find the approximation in each case as well as the error.
- c) Repeat the example with spline interpolation.

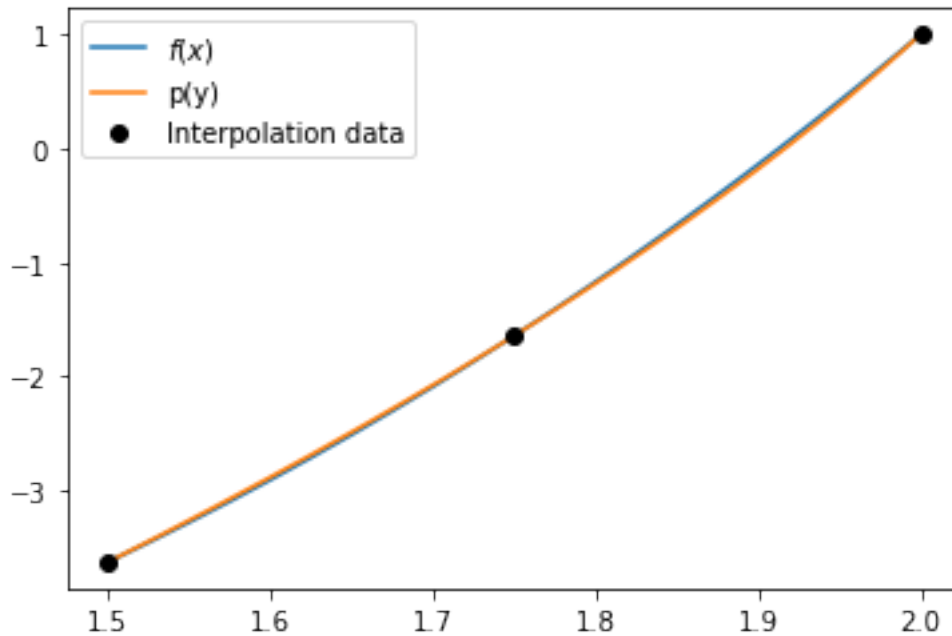
Solution.

- a) Interpolating in these (approximated) points gives the polynomial

$$1.91643 + 0.0903377y - 0.00676966y^2.$$

The polynomial evaluated in $y = 0$ is

$$p(0) = 1.91643 \Rightarrow |e_2| = 0.0035.$$

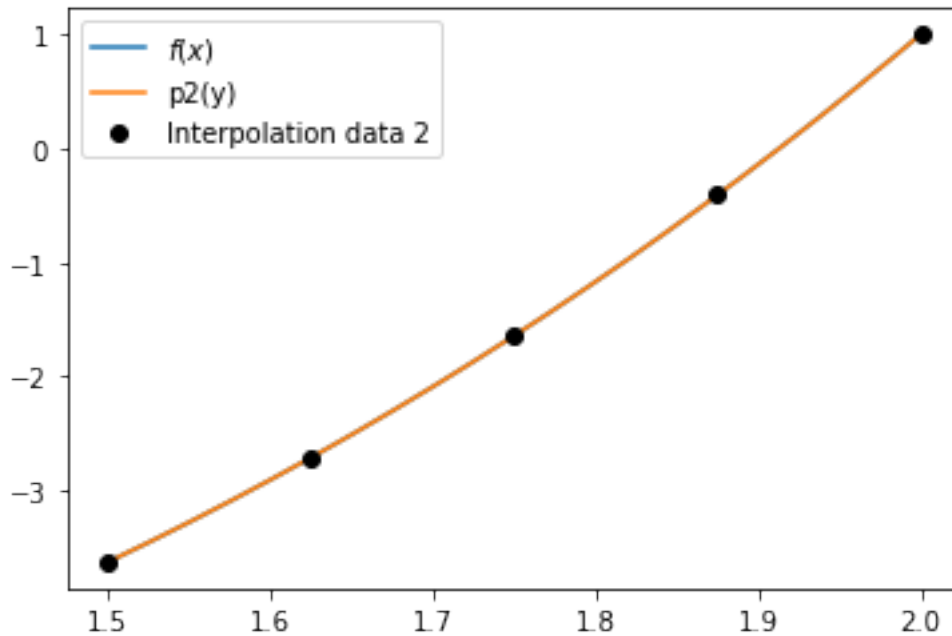


- b) Considering $n = 2$, there are three equidistant points that have been performed in a). For $n = 4$, the polynomial is

$$1.913 + 0.0912605y - 0.00436612y^2 + 0.000198267y^3 - 0.0000889468y^4.$$

The approximation is

$$p(0) = 1.913 \Rightarrow |e_2| = 0.0000688.$$

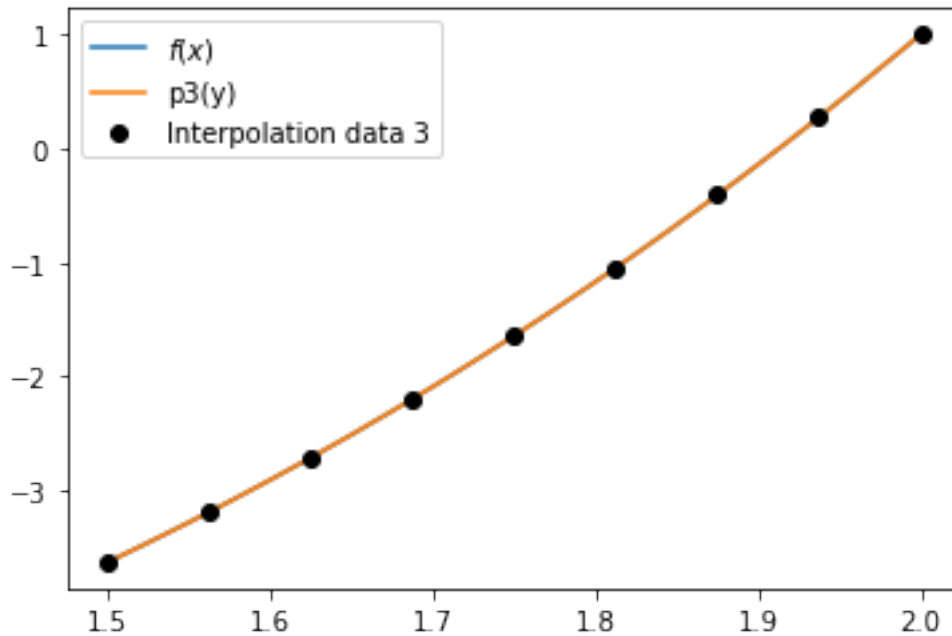


For $n = 8$, the polynomial is

$$1.91293 + 0.0910919y - 0.0043369y^2 + 0.0003455y^3 - 0.0000326y^4 + 0.0000024y^5 - 0.0000012y^6 - 0.0000002y^7 - 0.00000004y^8.$$

The approximation is

$$p(0) = 1.91293 \Rightarrow |e_2| = 0.00000118.$$

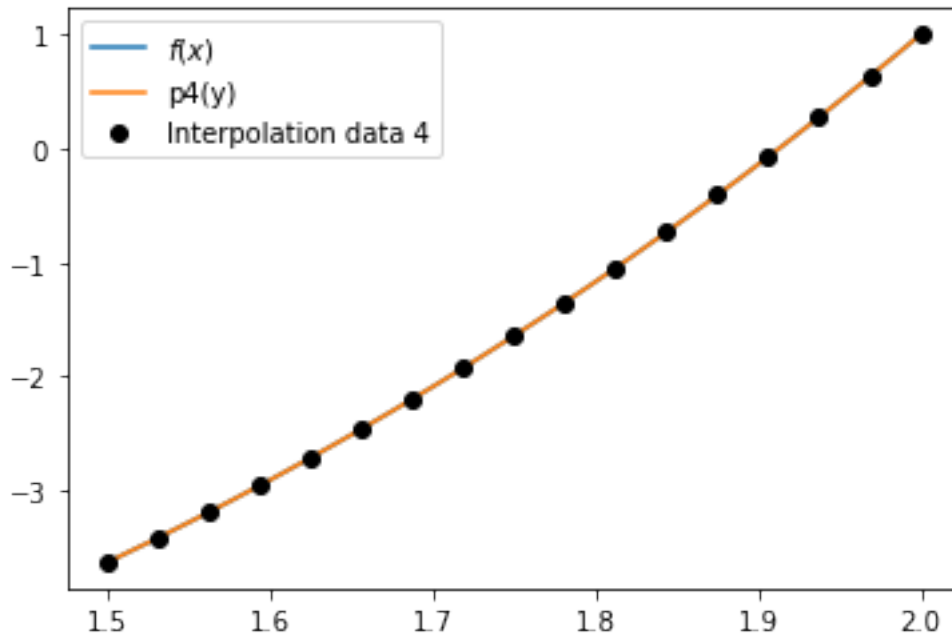


For $n = 16$, the polynomial is

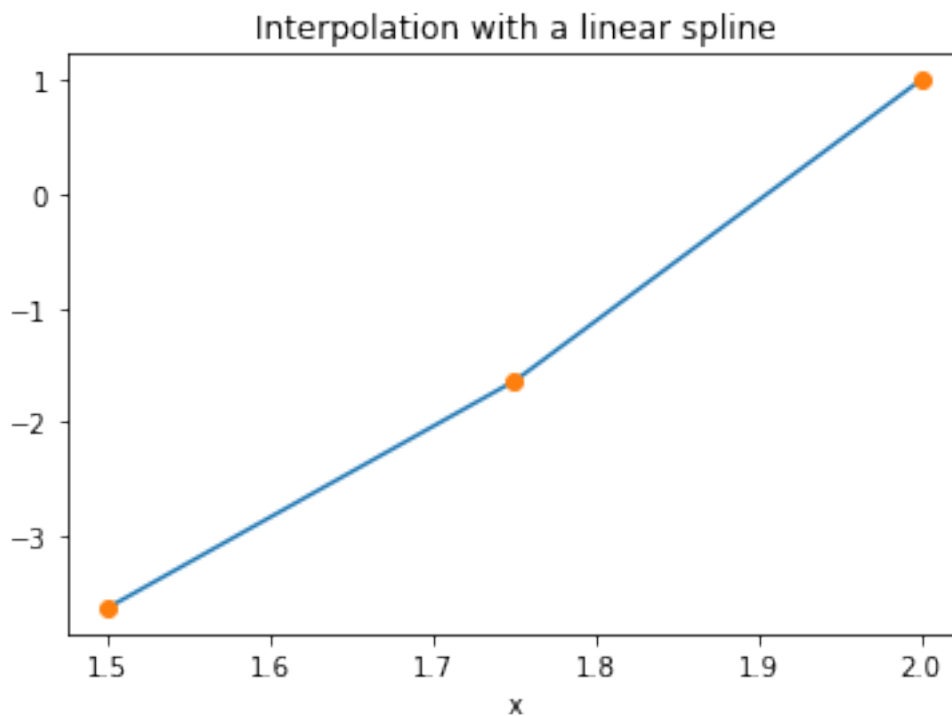
$$\begin{aligned}
 &1.91293 + 0.0910919y - 0.00433781y^2 + 0.000345199y^3 - 0.0000299598y^4 \\
 &+ 4.63397e - 6y^5 - 5.89262e - 6y^6 - 9.45272e - 6y^7 - 4.84843e - 6y^8 + 2.79734 - 6y^9 \\
 &+ 5.80398e - 6y^{10} + 4.16815e - 6y^{11} + 1.7502e - 6y^{12} + 4.65737e - 7y^{13} \\
 &+ 7.76611e - 8y^{14} + 7.43685e - 9y^{15} + 3.12993e - 10y^{16}.
 \end{aligned}$$

The approximation is

$$p(0) = 1.91293 \Rightarrow |e_2| = 0.00000118.$$



c) See *Exercise_2_P5.ipynb*



Problem 7. (Midpoint rule)

We consider in this problem the numerical integration of the function $f(x) = \log(x)$ on the interval $I = [0, 1]$.

- a) Implement the composite midpoint rule for the numerical computation of this integral.
- b) Estimate the numerical converge rate for this method. That is, assume that the error satisfies

$$\text{err}_m := \left| M_m(0, 1) - \int_0^1 \log(x) dx \right| \approx Ch^p,$$

and estimate numerically the number p .

Explain why your result from part b) does not contradict the usual theory of the midpoint rule.

Solution.

- a) The exact value of the integral is

$$\int_0^1 \log x dx = -1.$$

Using the numerical verification shown in Preliminaries,

$$p \approx \frac{\log(e(h_{m+1})/e(h_m))}{\log(h_{m+1}/h_m)}.$$

We choose a large number for m , for example, $m = 300$, then the value of p is approximately 0.9995.

- b) The value of p is different from 2. From the usual theory, we would expect that $p = 2$ if f is 2-times continuously differentiable. However, the function $\log(x)$ becomes to $-\infty$ at 0 and thus is not continuous, let alone (twice) differentiable.